

Name: Maths Class:

SYDNEY TECHNICAL HIGH SCHOOL



YEAR 11 Preliminary Course

Extension 1 Mathematics

Assessment 2

July 2013

TIME ALLOWED: 75 minutes

Instructions:

- Write your name and class at the top of this page, and on your answer booklet.
- Both this question sheet and your answer booklet must be handed in.
- All necessary working must be shown. Marks may not be awarded for careless or badly arranged work.
- Marks indicated within each question are a guide only and may be varied at the time of marking
- Approved calculators may be used.
- PART A is worth 5 marks, and is to be answered on the multiple choice answer sheet, which is the first page in your answer booklet. It should take about 7 minutes. **DO NOT DETACH IT.**
- PART B is to be written in the answer booklet provided. *Start each question on a new page.*

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PART A: (5 Marks)

Answers to these multiple choice questions are to be completed using the multiple choice answer sheet, which is the first page of your answer booklet

All questions are worth 1 mark

1	The function $f(x) = \frac{3x}{16+x^2}$ is: A. Odd B. Even C. Neither D. Cannot be determined
2	The line through the point (2, -5) and parallel to $3x - 2y = 6$ is: A. $3x - 2y - 16 = 0$ B. $3x + 2y + 4 = 0$ C. $2x - 3y - 19 = 0$ D. $2x + 3y + 11 = 0$
3	$\sin^2 x \cos^2 x =$ A. $\frac{1}{4} \sin^2 \frac{x}{2}$ B. $\frac{1}{4} \sin^2 2x$ C. $4 \sin^2 \frac{x}{2}$ D. $4 \sin^2 2x$
4	The solution to $\frac{2}{x-1} \leq 1$ is: A. $1 \leq x \leq 3$ B. $x \leq 1$ or $x \geq 3$ C. $1 < x \leq 3$ D. $x < 1$ or $x \geq 3$
5	If $f(x) = x^2 + 2$ and $g(x) = \sqrt{x-2}$ which of the following is true? A. $f\{g(x)\} = x$ B. $g\{f(x)\} = x$ C. Both are true D. Neither is true

PART B

(START EACH QUESTION ON A NEW PAGE)

QUESTION 6: (10 Marks)

	Marks
(a) Find the exact value of $\cos \frac{\pi}{6}$	1
(b) If $2\sin x = \sin(x + 90^\circ)$ find the value of $\tan x$	2
(c) (i) Fully factorise: $x^3 - 4x^2 - 16x + 64$	1
(ii) A curve is defined by $y \begin{cases} = \frac{x^3 - 4x^2 - 16x + 64}{x - 4}, & x \neq 4 \\ = 1, & x = 4 \end{cases}$	1
Sketch this curve	
(d) Find the acute angle between the lines $x - 2y = 1$ and $x = 2$, giving your answer to the nearest minute.	2
(e) If $\cos x = \frac{3}{5}$, $0 < x < 90^\circ$ and $\sin y = \frac{7}{25}$, $90^\circ < y < 180^\circ$ find the exact value of $\cos(x - y)$, giving your answer as a fraction.	3

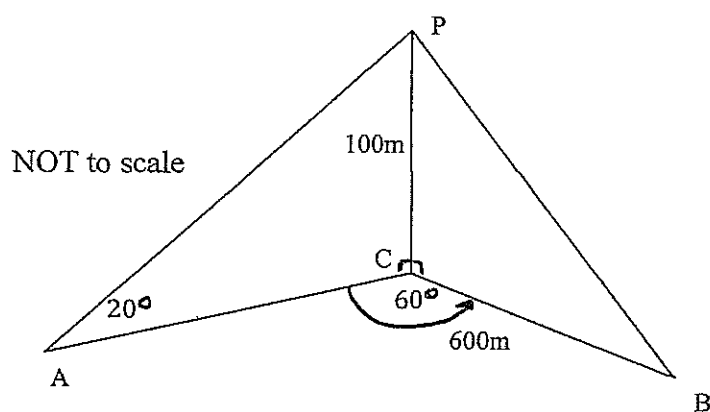
QUESTION 7: (9 Marks) Start a New Page**Marks**

(a) Solve the inequality $x - 1 > \frac{2}{x}$ 2

(b) Find the distance between the parallel lines: 2

$$2x - y + 2 = 0 \quad \text{and} \quad 2x - y = 5$$

(c)



Two yachts A and B subtend an angle of 60° at the base of a cliff, C.
From yacht A, the angle of elevation of the point P, 100m vertically above C, is 20°
Yacht B is 600m from C.

(i) Calculate the length AC to the nearest metre. 2

(ii) Calculate the distance between the two yachts, to the nearest metre 2

(d) Find $\frac{d}{dx} 3\sqrt{x}$ 1

QUESTION 8: (9 Marks) Start a New Page

Marks

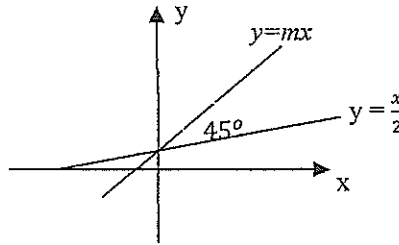
- (a) (i) M is the point which divides the interval joining A (-1, 3) to B (6, 17) in the ratio 5 : 2.
Find the co-ordinates of M. 1
- (ii) Find the equation of the line passing through M which is perpendicular to the line joining the points C (-1, 3) and D (5, 6) 3
- Give your answer in general form.
- (b) (i) Prove that $\sin(x + y) + \sin(x - y) = 2\sin x \cos y$ 1
- (ii) Hence find the exact value of $\sin 75^\circ + \sin 15^\circ$ 2
- (c) Find $\lim_{h \rightarrow 0} \left[\frac{(3+h)^2 - 9}{h} \right]$ 2

QUESTION 9: (9 Marks) Start a New Page

Marks

- (a) The angle between the 2 lines shown below is 45° .

2



Find the value of m .

- (b) Find the co-ordinates of the point which divides the line joining the points $(-2, 7)$ to $(1, 5)$ externally in the ratio $2:1$.

2

- (c) Shade the region which solves both of the following inequalities simultaneously:

4

$$(x - 2)^2 + (y + 2)^2 \leq 4 \quad \text{and} \quad x - y - 4 \leq 0$$

- (d) Differentiate with respect to x :

1

$$\frac{4x^3 - x}{x^2}$$

QUESTION 10: (9 Marks) Start a New Page

Marks

- (a) (i) Express $\sqrt{3}\cos x + \sin x$ in the form $R\cos(x - \alpha)$ where $0^\circ \leq \alpha \leq 90^\circ$ 2
- (ii) Hence, or otherwise, solve $\sqrt{3}\cos x + \sin x = 1$ for $0^\circ \leq x \leq 360^\circ$ 2
- (b) The circle $(x - 2)^2 + y^2 = 1$ has its centre at Q
- (i) Give the coordinates of Q 1
- (ii) Sketch the circle 1
- (iii) P is a point on the circle in the 1st quadrant so that the angle POQ is a maximum. (O is the origin). 1
What type of line is OP with respect to the circle?
- (iv) Find the size of the angle POQ. 2

QUESTION 11: (9 Marks) Start a new page.

Marks

(a) Differentiate $f(x) = x^2 + x$ using the method of First Principles. **3**

(b) Find the exact value of $\cos \frac{\pi}{12}$ using the expansion for $\cos 2x$ **2**

DO NOT EVALUATE THIS.

(c) (i) By finding their perpendicular distances to the origin, or otherwise, show that the lines $x + y = 1$ and $x + 7y - 5 = 0$ are both tangents to the same circle. **3**

(ii) Give the equation of this circle. **1**

SOLUTIONS — ASSESSMENT 2 — JULY.

EXTENSION 1

PART A

1/ A 2/ A 3/ B 4/ D 5/ C

PART B

QUESTION 6:

(a) $\sqrt{3}/2$

1 MARK

(b) $2 \sin x = \cos x$

1 for converting $\sin(x+90^\circ)$

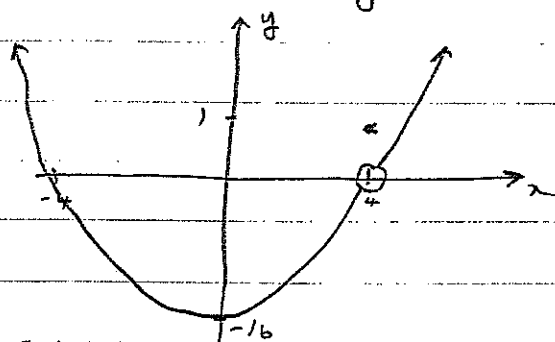
$\therefore \tan x = 1/2$

1 for answer.

(c) (i) $x^2(x-4) - 16(x-4)$
 $= (x-4)^2(x+4)$

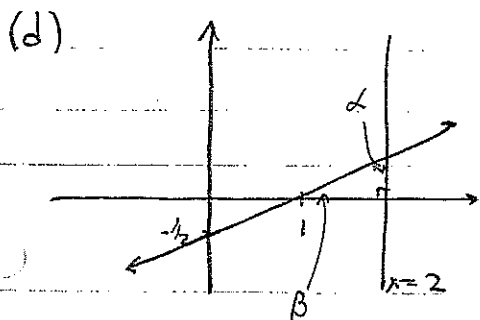
1 MARK or $(x-4)(x-4)(x+4)$

(ii) line becomes $y = (x-4)(x+4)$ except at $x=4$



1 for sketch.

(everything must be correct!)

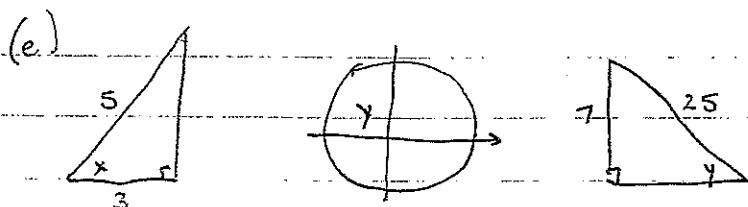


$\tan \beta = 1/2$
 $\beta = 26^\circ 34'$

1 for β

$\alpha = 90^\circ - \beta$
 $= 63^\circ 26'$

1 for α .



$\cos(x-y) = \cos x \cos y + \sin x \sin y$
 $= \frac{3}{5} \left(-\frac{24}{25}\right) + \frac{4}{5} \left(\frac{7}{25}\right)$
 $= \frac{-72+28}{125}$
 $= -\frac{44}{125}$

1 for $\cos y$ $\left(-\frac{24}{25}\right)$
 1 for $\sin x$ $\left(\frac{7}{25}\right)$

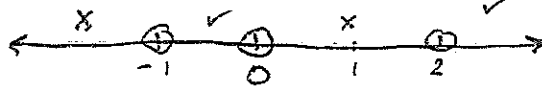
1 for answer.

QUESTION 7:

(a) C.V. $x \neq 0$

The equation is: $x^2 - x = 2$

$$(x-2)(x+1) = 0$$



$$-1 < x < 2 \text{ or } x > 2$$

1 for each region
= 2 marks.

(b) A point on $2x - y + 2 = 0$ is $(0, 2)$

$$P = \left| \frac{2 \cdot 0 - 1(2) + 2}{\sqrt{5}} \right|$$

$$= \frac{7}{\sqrt{5}} \text{ units}$$

1 for method

1 for answer.

(c) (i) $\frac{AC}{100} = \frac{1}{\tan 20^\circ}$

$$AC = 274.75 \text{ m}$$

← ①

← ①

(ii) $AB^2 = AC^2 + 600^2 - 2 \times 600 \times AC \cos 60^\circ$

$$\therefore AB^2 = (274.75)^2 + 360000 - 1200(274.75)(\frac{1}{2})$$

← ①

$$\therefore AB = 520.28 \text{ m}$$

← ①

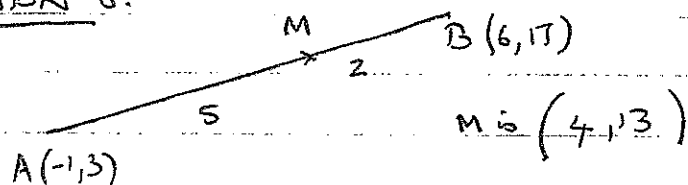
(d) $\frac{d}{dx} 3\sqrt{x} = 3(\frac{1}{2})x^{-\frac{1}{2}}$

$$= \frac{3}{2\sqrt{x}}$$

{ 1 for either but no penalty for getting the simplification wrong if the first answer is correct.

QUESTION 8:

(a)(i)



OR

$$M = \left(\frac{6 \cdot 5 + (-1) \cdot 2}{7}, \frac{17 \cdot 5 + 3 \cdot 2}{7} \right)$$

$$= (4, 13)$$

1 MARK for
answer, by
any method.

(ii) $m_{AB} = \frac{3}{6} = \frac{1}{2}$

$m_{\text{perpendicular}} = -2$

← 1 MARK

Line AB:

$$y - 13 = -2(x - 4)$$

$$2x + y - 21 = 0$$

② MARKS

(less 1 if not in general form)

(b) (i) $\sin(x+y) = \sin x \cos y + \cos x \sin y$
 $\sin(x-y) = \sin x \cos y - \cos x \sin y$

1 MARK

$$\sin(x+y) + \sin(x-y) = 2 \sin x \cos y$$

(ii) In the above, let $x = 45^\circ, y = 30^\circ$

1 MARK

$$\therefore \sin 75^\circ + \sin 15^\circ = 2 \sin 45^\circ \cos 30^\circ$$

$$= 2 \left(\frac{1}{\sqrt{2}} \right) \left(\frac{\sqrt{3}}{2} \right)$$

$$= \frac{\sqrt{3}}{\sqrt{2}} \left\{ \text{OR} \right.$$

$$\left. \frac{\sqrt{6}}{2} \right\}$$

1 MARK for either

(c) $\lim_{h \rightarrow 0} \left[\frac{(3+h)^2 - 9}{h} \right]$

$$= \lim_{h \rightarrow 0} \left[\frac{9 + 6h + h^2 - 9}{h} \right]$$

$$= \lim_{h \rightarrow 0} (6 + h)$$

$$= 6$$

① MARK for keeping the
"lim" until the end
 $h \rightarrow 0$
and in the correct spot
(ie not before the =)

← ①

QUESTION 10:

(a) (i) $R = 2$

$$\sqrt{3} \cos x + \sin x = 2 \left[\frac{\sqrt{3}}{2} \cos x + \frac{1}{2} \sin x \right]$$

$$= 2 \cos(x - \alpha)$$

where $\cos \alpha = \frac{\sqrt{3}}{2}$

$$\Rightarrow \alpha = 30^\circ \text{ (or } \frac{\pi}{6})$$

$$\therefore \sqrt{3} \cos x + \sin x = 2 \cos(x - 30^\circ)$$

(ii) $\sqrt{3} \cos x + \sin x = 1$

$$\therefore 2 \cos(x - 30^\circ) = 1$$

$$\therefore \cos(x - 30^\circ) = \frac{1}{2}$$

$$\therefore x - 30^\circ = 60^\circ \text{ OR } 300^\circ$$

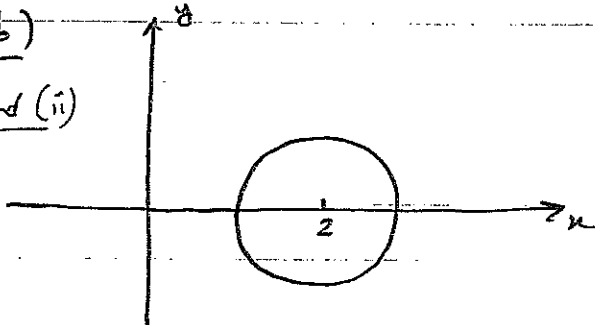
$$\therefore x = 90^\circ \text{ OR } x = 330^\circ$$

① for R

① for α

1 for each answer = ②

(b)
(i) and (ii)

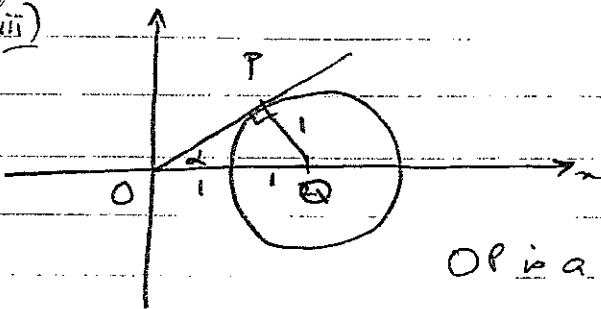


Q is (2, 0)

1 MARK

1 for sketch.

(iii)



OP is a tangent

1 MARK

Since $\angle OPQ = 90^\circ$, and $TQ = 1$, $OQ = 2$

$$\sin \alpha = \frac{1}{2}$$

$$\therefore \alpha = 30^\circ$$

} ① for solving all these.

← ① MARK

Question 11:

$$\begin{aligned}
 (a) \quad \frac{dy}{dx} &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{(x+h)^2 + (x+h) - (x^2 + x)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{2xh + h^2 + h}{h} \\
 &= \lim_{h \rightarrow 0} 2x + h + 1 \\
 &= 2x + 1
 \end{aligned}$$

3 MARKS

Look for:

- correct expansions
- keeping $\lim_{h \rightarrow 0}$ to the right of = sign.
- keeping the $\lim_{h \rightarrow 0}$ until they let $h = 0$.

[the answer is immaterial]

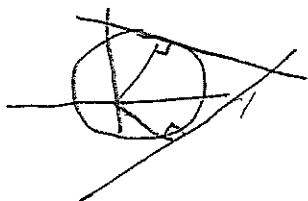
$$\begin{aligned}
 (b) (i) \quad \cos 2x &= 2\cos^2 x - 1 \\
 \therefore \cos \pi/6 &= 2\cos^2(\pi/12) - 1 \\
 \therefore \sqrt{3}/2 &= 2\cos^2(\pi/12) - 1 \\
 \therefore \cos^2 \pi/12 &= \frac{\sqrt{3}/2 + 1}{2} \\
 &= \frac{\sqrt{3} + 2}{4} \\
 \therefore \cos \pi/12 &= \sqrt{\frac{\sqrt{3} + 2}{4}}
 \end{aligned}$$

} 1 for realising this

1 for answer. Don't worry if they put $\pm$.

$$\begin{aligned}
 (c) (i) \quad p_1 &= \left| -\frac{1}{\sqrt{2}} \right| & p_2 &= \left| -\frac{5}{\sqrt{50}} \right| \\
 &= \frac{1}{\sqrt{2}} & &= \frac{1}{\sqrt{2}}
 \end{aligned}$$

$\therefore$ The perpendicular distances to the origin are the same. So they are both lines on the circumference of a circle and tangents to it



$$\begin{aligned}
 (ii) \quad r &= \frac{1}{\sqrt{2}}, \quad C: (0,0) \\
 x^2 + y^2 &= \frac{1}{2}
 \end{aligned}$$

3 MARKS for anything reasonably convincing.

1 MARK